Data Structures

BS (CS) _Fall_2024

Lab_09 Tasks



Learning Objectives:

1. Stack

Question 1

Make a stack class that implements all the following functions

1) Stack ()

    - Results: Constructor. Creates an empty stack.

2) ~Stack ()

    - Results: Destructor. Deallocates (frees) the memory used to store a stack.

3) void push (const DataItem)

    - Results: Push the element at the top of the stack.

4) int pop ()

    - Results: Returns the element from the top of the stack.

5) int Peek ()

    - Results: Return the element at the top of the stack.

6) void clear ()

    - Results: Removes all the elements from a stack.

7) Bool isEmpty ()

    - Results: Returns true if a stack is empty. Otherwise, returns false.

Question 2

Given a string s containing just the characters '(', ')', '{', '}', '[' and ']', determine if the input string is valid or not using stack.

An input string is valid if:

- Open brackets must be closed by the same type of brackets.
- Open brackets must be closed in the correct order.
- Every close bracket has a corresponding open bracket of the same type

Question 3

Implement a queue using two stacks